INCH-POUND

A-A-59131
08 OCTOBER 1997

COMMERCIAL ITEM DESCRIPTION

Detector Metal Flaw, Eddy Current, Impedance Plane, Lightweight

The General Services Administration has authorized the use of this Commercial item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description defines the minimum performance and technical requirements for a lightweight eddy current flaw detection instrument suitable for Nondestructive Inspection (NDI) use by U. S. Air Force field and depot NDI activities on ferrous and non-ferrous metals.

2. SALIENT CHARACTERISTICS.

2.1 General. The unit shall weigh less than 7 pounds, shall use an impedance plane display, shall have a dual variable frequency, and be capable of operating bolt hole scanners.

2.2 Size and weight. The unit shall not exceed 11 inches by 6 inches by 6 inches. The weight of the unit shall not exceed 7 pounds with batteries installed.

2.3 Power requirements. The unit shall operate when supplied with either alternating current or batteries as described below.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: SA-ALC/TILDD, 485 Quentin Roosevelt Drive, Bldg. 171, Kelly AFB TX 78241-6425 Document Improvement Proposal(DD Form 1426) appearing at the end of this document or by letter..

AMSC N/A FSC 6635

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2.3.1 AC Power. The unit shall operate from standard 115/220 volt +/- 10%, 50 and 60 Hz power supplies. Voltage and frequency selection switching can be automatic or manual.

2.3.2 Batteries. The battery packs shall be rechargeable and provide at least six (6) hours of continuous operation at 68 degrees F. The unit must incorporate a visible means of indicating the charge condition of the batteries and shall provide a minimum 15 minutes warning when the battery needs recharging. The battery packs shall be commercially available from multiple independent vendors. The unit shall be provided with two (2) battery packs.

2.3.3 Battery charger. The battery charger shall be separate from the unit from the instrument and simultaneously capable of charging the batteries and supplying operating power to the instrument.

2.3.4. Battery belt pack. The unit shall include a belt pack which can hold the batteries and provide power via an umbilical in order to reduce the weight of the hand held portion.

2.4 Controls. The unit function controls shall be menu driven with on-screen help available. As a minimum, the unit shall accommodate control of the following functions.

2.4.1 Frequency. The operating frequency shall be selectable from 100 Hz to 4 MHz (minimum range). There shall be at least 50 selectable frequencies in each of the following ranges: 100 Hz to 1 kHz, 1 kHz to 10 kHz, 10 kHz to 100 kHz, and 100 kHz to 1 MHz. There shall be a minimum of 20 selectable frequencies from 1 MHz to 4 MHz. The unit shall be capable of dual frequency operation.

2.4.2 Gain. The unit shall be capable of at least 85 dB (in 0.5 dB steps) of amplification range for both the vertical (y) and horizontal (x) axes.

2.4.3 Probe drive. The unit probe drive circuit shall allow the equipment operator to vary the signal driving the probe independent of gain and frequency adjustment.

2.4.4 Phase rotation. The unit shall be able to rotate the eddy current impedance vector (as viewed on the display) through a full 360 degrees in 1 degree increments or smaller.

2.4.5 Null. The unit shall incorporate a null circuit which, when activated, shall relocate the balance point to the defined reference location on the impedance plane display

2.4.6 Filters. The unit shall be equipped with high pass, low pass, and/or bandpass filters. The low pass filter shall be selectable from 10 Hz to 500 Hz (minimum range) and the high pass filter shall be selectable from 0 to 200 Hz (minimum range).

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2.4.7 Alarm function. The unit shall incorporate an alarm (gate) system that will trigger an audible and/or a visual alarm when the test signal enters or leaves a designated alarm zone. The equipment operator shall be able to vary the size, position, and shape (if applicable) of the alarm zones. The alarm shall be usable in the impedance plane and sweep mode displays.

2.4.8 Programmable test procedures. The unit shall have the capability to program, store, and recall a minimum of 50 instrument test parameter settings. The operator shall be able to label each setup alphanumerically for ease of recall. Programmed instrument settings shall be adjustable by the equipment operator. The programmed settings shall not be lost when the main power supply or battery is removed/turned off for a period of 300 hours minimum.

2.5 Instrument display. The display shall have a minimum diagonal dimension of 3.5 inches with a minimum resolution of 2250 pixels per square inch. The screen shall have a square grid pattern to enable dimensioning of the test signals. The display shall be clearly readable in bright and low light conditions.

2.5.1 Persistence. The display persistence shall be variable and adjustable by the operator from 0.1 to 10 seconds (minimum range) or permanent.

2.5.2 Display modes. The unit shall display eddy current inspection signals in the impedance plane (flying dot), the vertical amplitude versus time (sweep), and dynamic modes.

2.5.2.1 Impedance plane mode. The impedance plane display shall plot a trace of the eddy current signal as an impedance vector representing the vector sum of the test coil inductive reactance and resistance. The display shall incorporate separate controls for horizontal and vertical positioning of the balance point anywhere on the display screen.

2.5.2.2 Sweep mode. The sweep mode shall display the vertical impedance plane constituent of the eddy current signal as a function of time. The sweep trace rate shall be variable from 0.1 to 4 seconds per sweep. The unit shall make provisions for triggering the sweep externally through provided connections.

2.5.2.3 Dynamic mode. The display shall have a dynamic mode that is compatible with the scanners defined in paragraph 2.7.2. The dynamic display shall automatically initiate a sweep trace at the start of each scanner probe rotation and use a waterfall type display to show consecutive sweeps or show only the current sweep while automatically erasing the previous sweep information. The full 360 degrees of rotation shall be displayed regardless of any nominal changes in rotational speed of the scanner probe.

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2.6 Conductivity and coating thickness measurement. The unit shall be capable of measuring electrical conductivity and non-conductive coating thickness as specified below. The unit shall come equipped with appropriate probes and cables for conductivity and thickness measurement. The probe(s) shall be replaceable in the field without the need for depot or contractor intervention.

2.6.1 Electrical conductivity. The unit shall be capable of measuring electrical conductivity of materials in the range from 1 to 65% IACS. The unit shall use an operating frequency of 60 kHz for conductivity measurement. The measurement accuracy for the range from 20 to 65% IACS shall be within +/- 0.5% IACS at 68 degrees F with liftoff up to 0.008 inch. The accuracy for the range from 1 to 65% IACS shall be within +/- 1.0% IACS from 32 to 100 degrees F.

2.6.1.1 Conductivity standards. The unit shall include a set of conductivity reference standards sufficient for calibrating the instrument over its entire range. Conductivity reference standards shall be certified accurate to within +/- 0.2% IACS of their stated value.

2.6.2 Non-conductive coating thickness. The unit shall be capable of measuring non-conductive coatings up to 0.025 inch thick. The measurement accuracy shall be within +/- 5% of the reading.

2.7 External connections. In addition to the connections mentioned above, the unit shall accommodate the following external connections.

2.7.1 Probe connectors/adapters. The unit shall be compatible with current Air Force absolute, reflection, and differential type probes using BNC-to-MicroDot connector cables. Appropriate adapters shall be provided if required to connect to the BNC-equipped cables. Probes shall have temperature circuit included in the unit to compensate for probe temperature changes.

2.7.2 Scanner interface. The unit shall interface and be compatible with high speed and indexing bolt hole scanners. As a minimum, the eddy current unit shall provide the following: (1) scanner power, (2) sweep coordination (external triggering), (3) control of scanner probe revolutions per minute (RPM), (4) compatibility with remote null switching on the scanner (5) compensation for scanner speed variations due to varying hole tightness, and (6) capability to change probes in the field. The scanner control capability shall be included with the unit and be ready for use.

2.7.3 Analog output. The unit shall have analog outputs to allow for horizontal and vertical component test signal monitoring.

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2.7.4 PC interface. The unit shall include Microsoft Windows compatible software and required hardware to enable communication with a personal computer. Software capabilities shall include uploading and downloading instrument program settings and data reporting utilities such as downloading screen captures.

2.7.5 Printer output. The unit shall be capable of printing via an RS-232 serial port. The unit shall come with drivers for common commercially available serial printers.

2.8. Protective carrying cases. The unit shall include soft-shell and hard-shell carrying cases. The soft-shell case shall protect the unit from damage during use while still allowing access to all external connections. The hard-shell case shall be suitable for storing and shipping the unit with accessories.

2.9 Harness. The unit shall include an adjustable harness for "hands free" support of the unit in use.

2.10 Operating conditions. The unit shall be operable in the following environmental conditions.

2.10.1 Explosive atmosphere. The unit shall be tested and certified by a reputable laboratory for safe use, while running on batteries, in an explosive atmosphere as defined by Class 1, Division 2, Group D, in the National Fire Protection Association Code (NFPA 70E), Section 500. The testing method shall be in accordance with MIL-STD-810E, Method 511.3, Procedure 1. 2.9.2

2.10.2 Operating temperature. The unit and accessories shall perform to the specifications in this CID throughout temperatures from 0 to 120 degrees F at 95% relative humidity.

2.10.3 Storage temperature. The unit and accessories shall perform to the specifications in this CID following extended exposure to temperatures from 0 to 150 degrees F at 95% relative humidity.

2.10.4 Altitude. The unit and accessories shall perform to the specifications in this CID at altitudes from sea level to 7,280 feet above sea level, and following return from an altitude of 40,000 feet above sea level.

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3. REGULATORY REQUIREMENTS.

3.1 Recoverable materials. The contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

3.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description (CID), and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require a government witnessed, pre-award, sample verification test to insure conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 Market acceptance criteria. The company producing the item must have produced an eddy current instrument or similar instruments that is capable of meeting the requirements of this CID for at least 1 year, and have sold at least 100 eddy current instruments of the model being offered.

5. PACKAGING.

5.1 Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Ordering Data.

a) Title, number, and date of this description.
b) Specify special packaging requirements.
c) Contracting officer may request proof of certification of commercial item prior to first contract delivery.

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6.2 Component list. The following minimum components shall be supplied with the eddy current unit:

a. Eddy Current Instrument, 1 each
b. Battery Pack, 2 sets
c. Battery Charger/Eliminator (with cables), 1 each
d. Battery Belt Pack, 1 each
e. Conductivity Probe and Cable, 1 each
f. Conductivity Reference Standards, 1 set
g. Probe Cable Adapters, 1 each, as required
h. Soft-Shell Carrying Case, 1 each
i. Hard-Shell Shipping/Carrying Case, 1 each
j. Harness, Adjustable, 1 each
k. Operation Manual in English, 1 each
l. Familiarization Video, VHS format, 1 each
m. PC Interface Software/Accessories, (optional)

6.3 National Stock Number(s). The applicable NSNs for this test set are:

NSN	REFERENCE NO.	CAGEC
6635-01-00463-1574	CID97WL/MLS-0L6635-0178	98750

6.4 Sources of documents.

6.4.1 Federal Standards may be obtained from the General Services Administration, Federal Supply Service, Office of Commodity Management, Washington, DC, 20406.

6.4.2 Federal Acquisition Regulations may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.4.3 Electronic Industries Association(EIA) document may be obtained from EIA, Administration Dept., 2001 I ST N W, Washington, DC, 20006-1807.

MILITARY INTEREST:

CIVIL AGENCY COORDINATING ACTIVITY: GSA-FSS

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CUSTODIANS:
Air Force-99
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PREPARING ACTIVITY:
AIR FORCE 82

Review Activities:
DLA - GS

AGENT:
AIR FORCE-99

PROJECT NUMBER: 6635-0178

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